

Counting and Combinatorics

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Introduction

This lesson develops the basic toolkit of finite combinatorics: permutations, combinations, multinomial coefficients, the pigeonhole principle, and inclusion–exclusion. The centerpiece is the binomial theorem, which we state and prove by induction. Combinatorial reasoning underlies all of discrete probability and many counting arguments in machine learning theory (sample-space sizes, VC growth functions, counting of hypothesis classes).

Basic Principles

Definition 1 (Multiplication Principle). If a procedure consists of k stages, with n_i choices available at stage i independently of previous choices, then the total number of ways to execute the procedure is $\prod_{i=1}^k n_i$.

Definition 2 (Addition Principle). If a set S is the disjoint union $S = \bigsqcup_{i=1}^k S_i$, then $|S| = \sum_{i=1}^k |S_i|$.

Permutations and Combinations

Definition 3 (Permutation). Let n, k be nonnegative integers with $k \leq n$. A k -permutation of an n -element set S is an injection $\{1, \dots, k\} \rightarrow S$, equivalently an ordered tuple of k distinct elements of S . The number of k -permutations is

$$P(n, k) = \frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1).$$

Definition 4 (Combination). A k -combination of an n -element set S is a k -element subset of S . The number of k -combinations is the binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Proposition 1 (Pascal's rule). *For $1 \leq k \leq n - 1$,*

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. Fix an element $x \in S$ with $|S| = n$. A k -subset either contains x (there are $\binom{n-1}{k-1}$ such, choosing the remaining $k-1$ from $S \setminus \{x\}$) or excludes x (there are $\binom{n-1}{k}$). These cases are disjoint, so the count is the sum. $\square$

The Pigeonhole Principle

Definition 5 (Pigeonhole Principle). If $f: X \rightarrow Y$ is a function between finite sets with $|X| > |Y|$, then f is not injective: there exist $x_1 \neq x_2$ in X with $f(x_1) = f(x_2)$. More generally, if $|X| > k \cdot |Y|$, then some $y \in Y$ has at least $k + 1$ preimages.

The Binomial Theorem

Theorem 2 (Binomial Theorem). *For all $n \in \mathbb{Z}_{\geq 0}$ and all x, y in a commutative ring,*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Proof. We proceed by induction on n .

Base case ($n = 0$): Both sides equal 1. Indeed $(x + y)^0 = 1$ and $\sum_{k=0}^0 \binom{0}{k} x^k y^{-k}$ has its single term $\binom{0}{0} x^0 y^0 = 1$.

Inductive step: Assume the formula holds for some $n \geq 0$. Then

$$\begin{aligned} (x + y)^{n+1} &= (x + y)(x + y)^n = (x + y) \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} x^{k+1} y^{n-k} + \sum_{k=0}^n \binom{n}{k} x^k y^{n-k+1}. \end{aligned}$$

Reindex the first sum with $j = k + 1$:

$$\sum_{k=0}^n \binom{n}{k} x^{k+1} y^{n-k} = \sum_{j=1}^{n+1} \binom{n}{j-1} x^j y^{n+1-j}.$$

The second sum is already indexed in the form $\sum_{j=0}^n \binom{n}{j} x^j y^{n+1-j}$. Combining and isolating the $j = 0$ and $j = n + 1$ boundary terms,

$$\begin{aligned} (x + y)^{n+1} &= \binom{n}{n} x^{n+1} y^0 + \sum_{j=1}^n \left[\binom{n}{j-1} + \binom{n}{j} \right] x^j y^{n+1-j} + \binom{n}{0} x^0 y^{n+1} \\ &= x^{n+1} + \sum_{j=1}^n \binom{n+1}{j} x^j y^{n+1-j} + y^{n+1} \end{aligned}$$

by Pascal's rule. Since $\binom{n+1}{0} = \binom{n+1}{n+1} = 1$, this is exactly $\sum_{j=0}^{n+1} \binom{n+1}{j} x^j y^{n+1-j}$, completing the induction. $\square$

Inclusion–Exclusion

Theorem 3 (Inclusion–Exclusion). *For finite sets $A_1, \dots, A_n$,*

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

Worked Examples

(a) Permutation. The number of ways to arrange 3 chosen books in order out of 7 available is $P(7, 3) = 7 \cdot 6 \cdot 5 = 210$.

(b) Combination. The number of 5-card poker hands from a standard deck is $\binom{52}{5} = 2,598,960$.

(c) Pigeonhole. Among any 367 people, at least two share a birthday: $|X| = 367 > 366 = |Y|$.

(d) Inclusion–Exclusion. Integers in $\{1, \dots, 100\}$ divisible by 2, 3, or 5:

$$50 + 33 + 20 - 16 - 10 - 6 + 3 = 74.$$

(e) Binomial theorem. Setting $x = y = 1$ in Theorem 2 yields $2^n = \sum_{k=0}^n \binom{n}{k}$, the count of all subsets of an n -set.